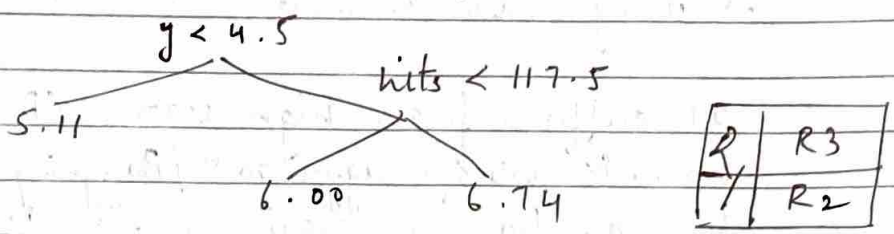
FABER-CASTELL
Date
Page No.

Lecture 32-33
Tree-Based Methods

- Tree works well for reg. & classificaⁿ. It involves stratifying & segmenting the predictor space into a no. of simple reg. It uses mean/median from training obs. of that region for predicⁿ. They can be summarized in a form of tree, hence, DT.



R_1, R_2, R_3 → terminal nodes / leaves

- 2 steps
1. divide the predictor space → $X_1 \dots X_p$ into J distinct & non-overlap reg. $R_1 \dots R_J$

$$\min \text{RSS} \quad \sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

we use greedy → recursive binary splitting

- It leads to complex trees that overfit. Smaller trees will have lower var. & better interpretaⁿ but ↑ bias. So we can have a large tree & then prune it.
- α using CV → non-negative HPT

$$\alpha \rightarrow T \subset T_0 \text{ s.t.} \quad \sum_{m=1}^{|T|} \sum_{x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

- In classificaⁿ, mini. Gini Index

$$G = \sum_{m=1}^{K} \hat{p}_{mk} (1 - \hat{p}_{mk})$$

$\hat{p}_{mk}$ → proporⁿ of training obs. in m^{th} reg. from k^{th} class.

node purity ← measure of total var. across K class.

↳ value, node has obs. from a single class.

- Advan.
  → easy to explain
  → mirrors decision-making by human
  → easy to interpret
  → handles qualitative predictor w/o dummy var..
- Disadv.
  → not that good predictive accuracy
  → very non-robust.

- DT suffers from high var. CR have low var. Bootstrap aggrega^n / Bagging is a method to red. bagging var. of stat. learning.
- Bagging → have B reg. trees from B bootstrapped samples & avg. the predictions.
  → reduces var., reduces interpretability
- RF → better than bagging, each time a split in a tree is considered, a random sample of m predictors is chosen as split candidates from the full set of p predictors.
  $m = \sqrt{p}$
  → decorrelates the trees → making the avg. of trees less variable & more reliable.